

Topic I: Trees – Terminology & Traversal Foundations

Binary Trees, Degree Bounds, Leaf Relations, Depth vs. Height, Traversal Properties

Section 1: Previous Year Questions (PYQs 1 to 3)

Q1. Leaf Nodes vs Degree-2 Nodes in Strict Binary Trees

GATE CS 2002 / DA Sample | MCQ

In any non-empty binary tree T where n_0 denotes the number of leaf nodes and n_2 denotes the number of internal nodes with degree 2, which of the following equations strictly holds?

(A) $n_0 = n_2 + 1$

(B) $n_0 = 2n_2$

(C) $n_0 = n_2 - 1$

(D) $n_0 = \lceil n_2 / 2 \rceil$

Answer: (A) Step-by-step Solution: Total nodes $n = n_0 + n_1 + n_2$. Total incoming edges is $n - 1$, which equals total outgoing degrees: $e = 0 \cdot n_0 + 1 \cdot n_1 + 2 \cdot n_2 = n_1 + 2n_2$. Equating both expressions: $n_0 + n_1 + n_2 - 1 = n_1 + 2n_2 \Rightarrow n_0 = n_2 + 1$.

Q2. Null Pointers in an n -node Binary Tree

GATE CS 2005 | NAT

A binary tree contains exactly 25 distinct nodes, with each node implemented with two child pointer fields (left and right). How many null pointers exist in this tree representation?

Answer: 26 Step-by-step Solution: For any binary tree with n nodes, the total number of child pointers is $2n$. Exactly $(n - 1)$ edges connect the parent nodes to non-root children (non-null pointers). Therefore, the number of null pointers is $2n - (n - 1) = n + 1$. For $n = 25$, null pointers = $25 + 1 = 26$.

Q3. Height and Maximum Leaves in Binary Trees

GATE CS 2015 | MCQ

Let the height of a single-node tree be defined as 0. What is the maximum possible number of leaf nodes in a binary tree of height h ?

(A) 2^h

(B) $2^{h+1} - 1$

(C) 2^{h-1}

(D) $2^h - 1$

Answer: (A) Step-by-step Solution: At depth d , a binary tree can have at most 2^d nodes. In a tree of height h , all leaves can be situated at the maximum depth level h . Thus, the maximum number of leaves is 2^h .

Q4. Inorder Sort Invariance in Binary Search Trees

GATE DA 2024 / CS 2014 | MCQ

Which of the following traversals on a Binary Search Tree (BST) visits the stored keys in strictly ascending sorted order?

- (A) Preorder traversal
- (B) Inorder traversal
- (C) Postorder traversal
- (D) Level-order traversal

Answer: (B) Step-by-step Solution: By BST definition, for any node x , all keys in the left subtree are $< \text{key}(x)$ and all keys in the right subtree are $> \text{key}(x)$. Inorder traversal processes (Left, Root, Right), which recursively prints keys in strictly ascending numerical order.

Q5. Preorder and Postorder Equality Condition

GATE CS 2011 | MCQ

If the preorder traversal sequence and postorder traversal sequence of a non-empty binary tree are identical, then the tree must have:

- (A) Exactly one node
- (B) Only left children
- (C) Only right children
- (D) Height equal to number of nodes

Answer: (A) Step-by-step Solution: Preorder visits [Root, Left, Right] while postorder visits [Left, Right, Root]. The root is printed first in preorder and last in postorder. For the sequences to be identical, the root must be the first and only element. Hence, the tree consists of exactly 1 node.

Q6. Traversal Stack Aux-Space Bound

New Practice | MSQ

Which of the following statements is/are **TRUE** regarding binary tree traversals on a tree with n nodes and height h ? [Select all that apply]

- (A) Standard recursive inorder traversal requires $O(h)$ auxiliary call-stack space.
- (B) Level-order traversal using a queue requires $O(2^h)$ worst-case space.
- (C) Morris traversal achieves inorder traversal in $O(n)$ time and $O(1)$ auxiliary space.
- (D) Preorder and Postorder uniquely determine an arbitrary binary tree.

Answer: (A), (B), and (C) Step-by-step Solution: (A) Stack depth matches path length = $O(h)$. (B) The maximum queue width occurs at the last level of a full tree = $O(2^h)$. (C) Morris traversal uses threaded binary tree links for $O(1)$ space. (D) is false: Preorder + Postorder does not uniquely determine a binary tree if any internal node has only one child.

Q7. Postorder Traversal Trace from BST Preorder

GATE CS 2017 | MCQ

The preorder traversal of a Binary Search Tree (BST) is: [30, 20, 10, 15, 25, 40, 50, 45]. Which of the following is the postorder traversal of this BST?

- (A) [15, 10, 25, 20, 45, 50, 40, 30] (B) [10, 15, 25, 20, 45, 40, 50, 30]
(C) [15, 20, 10, 25, 50, 45, 40, 30] (D) [15, 10, 25, 20, 45, 40, 50, 30]

Answer: (A) Step-by-step Solution: Root is 30. Left subtree values are < 30 : {20, 10, 15, 25}; Right subtree values are > 30 : {40, 50, 45}. For Left subtree: root is 20, left is 10 (with right child 15), right is 25. Postorder of left: 15, 10, 25, 20. For Right subtree: root is 40, right is 50 (with left child 45). Postorder of right: 45, 50, 40. Full postorder: [15, 10, 25, 20, 45, 50, 40, 30].

Q8. Traversal Output Value Computation

GATE CS 2008 | NAT

Consider a full binary tree with root 1. Each parent i has left child $2i$ and right child $2i + 1$ for indices up to 7. If the nodes are visited in Postorder traversal, what is the 5-th visited node index?

Answer: 6 Step-by-step Solution: The tree nodes are: root 1; left branch 2 (children 4, 5); right branch 3 (children 6, 7). Postorder visits: Left subtree of 1: [4, 5, 2]. Right subtree of 1: left child of 3 is 6, right is 7, then 3: [6, 7, 3]. Complete sequence: 4, 5, 2, 6, 7, 3, 1. The 1st is 4, 2nd is 5, 3rd is 2, 4th is 6? Wait: 1st=4, 2nd=5, 3rd=2, 4th=6, 5th is node 6? Wait: list is: 1st:4, 2nd:5, 3rd:2, 4th:6, 5th:7. Let's re-verify: 4, 5, 2 (3 nodes); 4th is 6, 5th is 7. Correction: 5-th node is 7.

Q9. Expression Tree Evaluation via Postorder Trace

New Practice | NAT

An arithmetic expression tree has root operator '+'. The left subtree evaluates the subexpression $(4 * (5 - 2))$ and the right subtree evaluates $(8 / (1 + 1))$. What integer value is returned at the completion of postorder traversal?

Answer: 16 Step-by-step Solution: Postorder evaluates children before the parent operator. Left operand: $5 - 2 = 3$; $4 * 3 = 12$. Right operand: $1 + 1 = 2$; $8 / 2 = 4$. Root operator '+': $12 + 4 = 16$.

Q10. Arc Consistency Definition (AC-3 Invariant)

GATE DA 2024 / AI Core | MCQ

A directed arc (X_i, X_j) in a constraint network is defined as **arc-consistent** if and only if:

- (A) For every value $x \in D_i$, there exists at least one value $y \in D_j$ satisfying the binary constraint. (B) For every value $y \in D_j$, there exists at least one value $x \in D_i$ satisfying the binary constraint.
(C) The domains D_i and D_j are disjoint. (D) All assignments to X_i and X_j satisfy every global constraint in the network.

Answer: (A) **Step-by-step Solution:** By standard Russell & Norvig AI definition, an arc (X_i, X_j) is arc-consistent with respect to constraint C_{ij} if for every value x in the current domain D_i of X_i , there is some value y in domain D_j of X_j such that the binary constraint between X_i and X_j is satisfied.

Q11. Worst-Case Time Complexity of the AC-3 Algorithm

GATE AI / CS Core | MCQ

In a binary CSP with n variables, each having maximum domain size d , and c binary constraints, the worst-case time complexity of the AC-3 algorithm is:

- (A) $O(c \cdot d^3)$ (B) $O(c \cdot d^2)$
(C) $O(n \cdot d^2)$ (D) $O(n^2 \cdot d)$

Answer: (A) **Step-by-step Solution:** A binary constraint network has $2c$ directed arcs. Each arc (X_i, X_j) can be inserted into the AC-3 worklist at most d times (since domain D_i has size d and can shrink at most d times). Testing an arc takes $O(d^2)$ comparisons. Thus, worst-case time is bounded by $(2c \cdot d) \times O(d^2) = O(c \cdot d^3)$.

Q12. Forward Checking vs Arc Consistency

GATE DA Model / AI Core | MSQ

Which of the following statements is/are **CORRECT** comparing Forward Checking (FC) and Arc Consistency (AC-3)? [Select all that apply]

- (A) Forward Checking only enforces arc consistency on arcs pointing from unassigned variables to the current assigned variable. (B) AC-3 propagates domain reductions transitively to other unassigned variables.
(C) Forward Checking detects all upcoming dead-ends earlier than AC-3. (D) Maintaining Arc Consistency (MAC) executes AC-3 after every assignment step.

Answer: (A), (B), and (D) **Step-by-step Solution:** FC checks arcs directed from neighbors to the newly assigned variable; it does not look ahead among unassigned variables. AC-3 propagates cascading reductions across the entire graph. Hence, AC-3 detects conflicts strictly earlier than FC (making C false). (D) is the precise definition of MAC.

Q13. AC-3 Domain Pruning Trace

[New Practice | NAT](#)

Consider variables X and Y with domains $D_X = \{1, 2, 3, 4\}$ and $D_Y = \{1, 2, 3, 4\}$. The constraint is $X + Y = 4$. After enforcing arc consistency on arc (X, Y) , what is the size of domain D_X ?

Answer: 3 Step-by-step Solution: For each $x \in D_X$, check if there is $y \in D_Y$ such that $x + y = 4$: For $x = 1$: $y = 3 \in D_Y$ (valid). For $x = 2$: $y = 2 \in D_Y$ (valid). For $x = 3$: $y = 1 \in D_Y$ (valid). For $x = 4$: requires $y = 0$, but $0 \notin D_Y$ (invalid). Value 4 is pruned. Remaining domain $D_X = \{1, 2, 3\}$, size = 3.

Q14. Backtracking Search Variable Ordering Heuristic

[New Practice | MCQ](#)

In Backtracking Search for CSPs, using the *Minimum Remaining Values (MRV)* heuristic is also commonly known as the:

- | | |
|--|--|
| (A) Most constrained variable heuristic (fail-first principle) | (B) Least constraining value heuristic |
| (C) Arc-inconsistency elimination heuristic | (D) Random restart heuristic |

Answer: (A) Step-by-step Solution: MRV selects the variable with the fewest legal values in its domain. Choosing the most constrained variable causes early failure and prunes unviable subtrees quickly, embodying the standard "fail-first" principle.

Q15. Tree-Structured CSP Direct Solvability

[New Practice | MCQ](#)

If the constraint graph of a discrete CSP with n variables and maximum domain size d contains no cycles (is tree-structured), it can be solved with **zero backtracking** in time:

- | | |
|------------------------|-------------------|
| (A) $O(n \cdot d^2)$ | (B) $O(d^n)$ |
| (C) $O(n^2 \cdot d^3)$ | (D) $O(n \log d)$ |

Answer: (A) Step-by-step Solution: Topologically sort nodes from root to leaves. Perform arc-consistency backwards from leaves to root ($n - 1$ arcs, each $O(d^2)$), then assign values forward from root to leaves without any conflicts or backtracking. Total time is strictly $O(n \cdot d^2)$.

Topic IV: Tree Reconstruction from Dual Traversals (PYQs 16-18)

Q16. Tree Reconstruction: Inorder & Preorder Sequences

GATE CS 2004 / 2015 | MCQ

A binary tree has traversals:

Preorder: P, Q, R, S, T, U, V | **Inorder:** Q, P, S, T, R, V, U.

Which of the following is the **Postorder** traversal of this tree?

(A) Q, T, S, V, U, R, P

(B) Q, T, S, U, V, R, P

(C) Q, S, T, V, U, R, P

(D) T, Q, S, V, U, R, P

Answer: (A) Step-by-step Solution: Root is P (first in Preorder). In Inorder, Q is to the left of P (left subtree has {Q}), while {S, T, R, V, U} is right subtree. In right subtree, next preorder root is R. In Inorder, {S, T} are left of R, and {V, U} are right of R. For {S, T}: preorder is S, T $\Rightarrow$ S is parent, T is right child. For {V, U}: preorder is U, V $\Rightarrow$ U is root, V is left child. Assembling postorder: Left child of P: Q. Right child of P: Left of R (T, S), Right of R (V, U), then R $\rightarrow$ Q, T, S, V, U, R, P.

Q17. Necessary Traversal Pairs for Unique Binary Tree Construction

GATE CS 2007 | MSQ

Which of the following pairs of traversals can uniquely reconstruct any arbitrary binary tree with distinct keys? [Select all that apply]

(A) Inorder and Preorder

(B) Inorder and Postorder

(C) Preorder and Postorder

(D) Inorder and Level-order

Answer: (A), (B), and (D) Step-by-step Solution: Inorder is strictly required to partition nodes into left and right subtrees. Combining Inorder with Preorder, Postorder, or Level-order uniquely determines the tree. Preorder and Postorder alone cannot differentiate whether a single child is a left or a right child (fails for trees with degree-1 internal nodes).

Q18. Height of Reconstructed Binary Tree

GATE CS 2014 | NAT

A binary tree with 6 nodes has:

Preorder: 1, 2, 3, 4, 5, 6 | **Inorder:** 2, 1, 4, 3, 6, 5.

Taking the height of a single-node tree as 0, what is the height of the reconstructed tree?

Answer: 3 Step-by-step Solution: Root is 1 (level 0). Inorder shows node 2 is in left subtree (level 1). Right subtree contains {4, 3, 6, 5}. Next preorder root is 3 (level 1). In Inorder, 4 is left child of 3 (level 2). Right subtree of 3 has {6, 5}. Preorder shows 5 is root (level 2), and Inorder shows 6 is left child of 5 (level 3). Maximum depth reached is at node 6 (level 3). Thus, height = 3.

Topic IV: Reconstruction Practice Questions (Practice 19 to 21)

Q19. Preorder from Inorder & Postorder Sequences

[New Practice | MCQ](#)

A binary tree with distinct elements has traversals:

Inorder: D, B, E, A, F, C | **Postorder:** D, E, B, F, C, A.

What is the Preorder traversal sequence of this binary tree?

(A) A, B, D, E, C, F

(B) A, B, E, D, C, F

(C) A, D, B, E, F, C

(D) A, B, D, E, F, C

Answer: (A) Step-by-step Solution: Root is A (last in Postorder). Inorder partitions: Left subtree = {D, B, E}, Right subtree = {F, C}. In Postorder, left subtree elements end with B, so B is the left child of A. In Inorder, D is left of B and E is right of B. In Postorder, right subtree elements are F, C, so C is root and F is its left child. Preorder (Root, Left, Right) is: A, B, D, E, C, F.

Q20. Full / Proper Binary Tree Reconstruction without Inorder

[New Practice | MCQ](#)

If a binary tree is known to be a **Full Binary Tree** (every node has either 0 or 2 children), it can be uniquely reconstructed from:

(A) Preorder and Postorder traversals alone

(B) Level-order traversal alone

(C) Preorder traversal alone

(D) Postorder traversal alone

Answer: (A) Step-by-step Solution: In a full binary tree, no node has only 1 child. In preorder [Root, Left, Right], the element immediately after Root is the root of the left subtree. Finding this element in postorder marks the end of the entire left subtree sequence. Thus, left and right subtrees are uniquely partitioned without ambiguity.

Q21. Rightmost Leaf Node Identification from Dual Traversals

[New Practice | NAT](#)

A binary tree has:

Preorder: 10, 5, 2, 7, 15, 12, 20 | **Inorder:** 2, 5, 7, 10, 12, 15, 20.

What is the numeric value of the rightmost node at the maximum depth of this tree?

Answer: 20 Step-by-step Solution: Notice that Inorder is strictly sorted: [2, 5, 7, 10, 12, 15, 20]. Therefore, this tree is a Binary Search Tree (BST). Root is 10. Right subtree of 10 has {12, 15, 20}. Preorder has 15 as root, 12 as left child, 20 as right child. Max depth in right subtree is reached at depth 2 with nodes 12 and 20. The rightmost node is 20.

Synthesis & Advanced Practice (Questions 22 to 24)

Q22. Postorder Reconstruction Recursive Call Count

[New Practice | NAT](#)

A divide-and-conquer function reconstructs a binary tree from Inorder and Preorder arrays of size 7. If every valid node instantiation requires one function invocation, how many successful recursive node-construction calls are made?

Answer: 7 Step-by-step Solution: Each recursive constructor call creates exactly one tree node from the preorder root and partitions the inorder boundary. In a tree with 7 distinct keys, exactly 7 node-construction calls are executed to create the 7 tree nodes.

Q23. Path Consistency (PC-2) vs Arc Consistency (AC-3)

[New Practice | MSQ](#)

Which of the following assertions regarding Higher-Level Consistency in CSPs is/are **CORRECT**? [Select all that apply]

- (A) A two-variable assignment $\{X_i = a, X_j = b\}$ is path-consistent with variable X_m if there exists $c \in D_m$ satisfying constraints with both.
- (B) Arc consistency implies path consistency in all networks.
- (C) A network that is k -consistent and $(k - 1)$ -consistent is strongly k -consistent.
- (D) Strongly n -consistent CSPs can be solved without backtracking.

Answer: (A), (C), and (D) Step-by-step Solution: (A) is the formal definition of path consistency. (C) and (D) are textbook properties: strongly n -consistent problems allow backtrack-free greedy assignment of all n variables. (B) is false: arc consistency (2-consistency) is strictly weaker than path consistency (3-consistency).

Q24. Leaf Node Count in Reconstructed Tree

[New Practice | NAT](#)

A binary tree with 7 distinct nodes has:

Preorder: A, B, D, E, C, F, G | **Inorder:** D, B, E, A, F, C, G.

What is the total number of leaf nodes in this tree?

Answer: 4 Step-by-step Solution: Root is A. Inorder partitions: Left = {D, B, E}, Right = {F, C, G}. In left subtree: root is B, with left child D and right child E. Both D and E are leaves. In right subtree: root is C, with left child F and right child G. Both F and G are leaves. Total leaf nodes = {D, E, F, G}, which equals 4 leaves.